Land Survey

Engineering Mathematics II for Land Survey

Professional Learner Notes

Coordinate geometry, whole-circle bearings, gradients, latitudes, departures and area from coordinates.

Section	Purpose
Coordinate geometry	Distance, gradient and line equations.
Whole-circle bearings	Direction measurement and quadrant interpretation.
Latitudes and departures	North-south and east-west components.
Gradients and slopes	Rise, run and percentage gradient.
Area from coordinates	Coordinate area method and plotting checks.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Gradient

$$\mathrm{Gradient}=\frac{\mathrm{Rise}}{\mathrm{Run}}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Departure

$$\Delta E=L\sin\theta$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Latitude

$$\Delta N=L\cos\theta$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Coordinate area

$$A=\frac{1}{2}\left|\sum x_iy_{i+1}-\sum y_ix_{i+1}\right|$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Distance

$$d=\sqrt{(\Delta E)^2+(\Delta N)^2}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Coordinate geometry

Distance, gradient and line equations.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\mathrm{Gradient}=\frac{\mathrm{Rise}}{\mathrm{Run}}$$

Worked method

Example: Traverse component. Find latitude and departure for a 100 m line at bearing 35°.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports levelling. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving control lines. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving area by coordinates. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving survey closures. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving levelling. Show formula, substitution, units and interpretation.

5. Create and solve one technical calculation involving traverse computation. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving control lines. Show formula, substitution, units and interpretation.

7. Create and solve one technical calculation involving area by coordinates. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving survey closures. Show formula, substitution, units and interpretation.

2. Whole-circle bearings

Direction measurement and quadrant interpretation.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

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\Delta E=L\sin\theta
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Worked method

Example: Gradient. Find gradient when level rises 1.5 m over 60 m horizontal distance.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports traverse computation. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving area by coordinates. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving survey closures. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving levelling. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving traverse computation. Show formula, substitution, units and interpretation.

5. Create and solve one technical calculation involving control lines. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving area by coordinates. Show formula, substitution, units and interpretation.

7. Create and solve one technical calculation involving survey closures. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving levelling. Show formula, substitution, units and interpretation.

3. Latitudes and departures

North-south and east-west components.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

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\Delta N=L\cos\theta
```

Worked method

Example: Area. Use coordinate method to find area of a small parcel.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports control lines. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving survey closures. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving levelling. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving traverse computation. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving control lines. Show formula, substitution, units and interpretation.

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4. Gradients and slopes

Rise, run and percentage gradient.

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Main formula for this section

$$A=\frac{1}{2}\left|\sum x_iy_{i+1}-\sum y_ix_{i+1}\right|$$

Worked method

Example: Traverse component. Find latitude and departure for a 100 m line at bearing 35°.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports area by coordinates. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

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Practice set

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5. Area from coordinates

Coordinate area method and plotting checks.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
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Mixed Revision Questions

1. A land survey problem involves traverse computation. Choose a suitable mathematical method, solve the problem, and explain the result.

2. A land survey problem involves control lines. Choose a suitable mathematical method, solve the problem, and explain the result.

3. A land survey problem involves area by coordinates. Choose a suitable mathematical method, solve the problem, and explain the result.

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27. A land survey problem involves control lines. Choose a suitable mathematical method, solve the problem, and explain the result.

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29. A land survey problem involves survey closures. Choose a suitable mathematical method, solve the problem, and explain the result.

30. A land survey problem involves levelling. Choose a suitable mathematical method, solve the problem, and explain the result.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.

2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.

3. Uploaded KNEC solved mathematics materials for examination-style orientation.

4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.